

Geometry

Unit 11

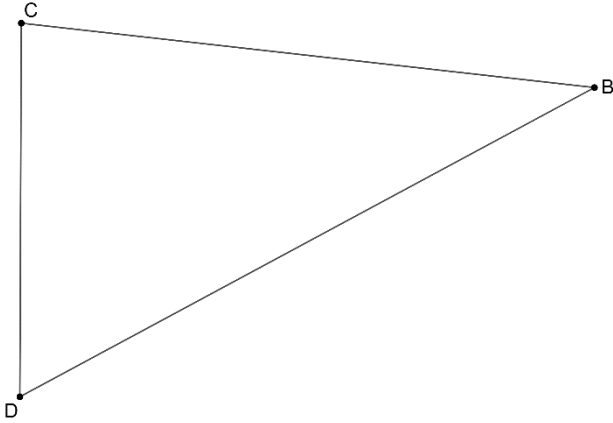
Lesson 7 Homework

Name _____

Date _____

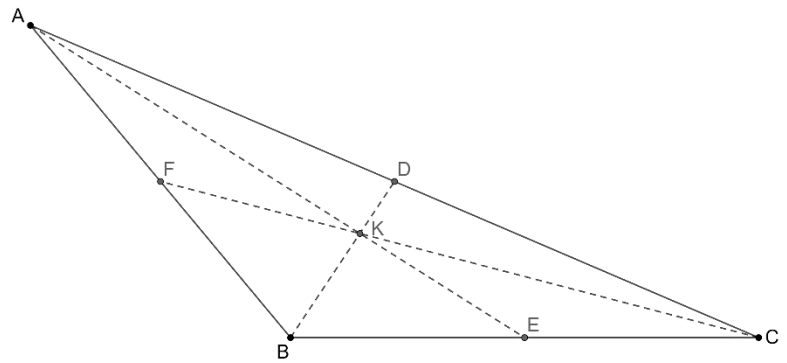
Period _____

$\triangle BCD$ is shown. Complete the steps for finding the centroid of the circle. Then, label the centroid Point Y .

Steps	Construction
Construct the perpendicular bisector of BC to find its midpoint.	
1.	
2.	5.
Draw the median, R , to the opposite vertex, $\angle DBC$.	
3.	
4.	
Draw Point Y , the point of concurrency, where the three medians intersect.	

Use this information to answer questions 6 through 8.

Triangle ABC is shown, where point K is the centroid of the triangle. $FC = 6$, $BK = 2.4$, and $KE = 3.7$.



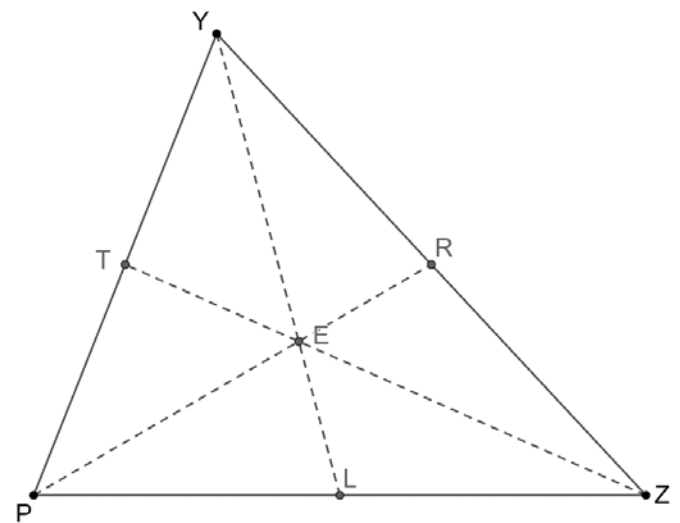
6. Find the length of $\overline{DK}$.

7. Find the length of $\overline{AE}$.

8. Find the length of $\overline{KC}$.

Use this information to answer questions 9 and 10.

Triangle YPZ is shown, where point E is the centroid of the triangle. $TZ = 6x + 9$, and $TE = 4$.



9. Find the value of x .

10. Find the length of $\overline{EZ}$.